

Retrieval-Augmented QA with Qwen2.5-1.5B and LoRA Fine-Tuning

Andrey Zhogov

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Abstract

This project develops a multilingual retrieval-augmented QA assistant that combines a lightweight 1.5 billion-parameter instruction model (Qwen2.5 1.5B Instruct) with a vector-store retriever built on FAISS. We specialize the model to an internal knowledge base in Russian using parameter-efficient fine-tuning (LoRA). Compared to the base model, LoRA produces a $3\times$ higher exact-match score ($33.6 \rightarrow 10.1$ pp gain) and +33 F1 on the XQuAD Russian dev set while reducing the verbosity of the answers. Project code: https://github.com/satisfied/faq_bot

1 Introduction

Modern enterprises need accurate conversational access to proprietary documents. Large language models alone hallucinate when queried about domain-specific content. Retrieval-augmented generation (RAG) mitigates this by grounding answers in retrieved passages. However, high-capacity models are costly; smaller open-weight models plus parameter-efficient fine-tuning offer an economical alternative. We investigate whether LoRA-tuned Qwen2.5-1.5B can reach useful accuracy on Russian QA while remaining deployable on CPU.

1.1 Team

Andrey Zhogov implemented the retrieval pipeline, LoRA training, evaluation, and report.

2 Related Work

[Lewis et al., 2020] RAG architecture (Lewis et al., 2020) integrates dense retrieval with sequence-to-sequence models.

[Hu et al., 2021] LoRA (Hu et al., 2021) injects low-rank adapters into attention weights, achieving strong task adaptation with $\leq 0.5\%$ extra parameters.

[Artetxe et al., 2020] XQuAD (Artetxe et al., 2020) extends SQuAD v1.1 to 10 languages, incl. Russian.

3 Model Description

Our system contains three layers:

1) Document Processor: directory crawler → recursive text splitter (1000 chars, 100 overlap).

2) Vector Store: LaBSE(LaBSE-en-ru) sentence embeddings indexed with FAISS; retrieval top-k=3.

3) Generator: Qwen2.5-1.5B-Instruct with merged LoRA adapters (rank=8, a=16). The prompt template wraps the retrieved context and user question in the Qwen chat schema.

The inference pipeline is exposed through a Bot class with a single run() method.

4 Dataset

Knowledge Base: 50k samples from SberQuAD dataset.

Evaluation: Russian portion (880 samples) of XQuAD dev set.

	Train	Valid	Test
QAs with context	50k	1000	880
Tokens	10M	224k	189k
Vocabulary size	151,936		
Out of Vocab rate	0%		

Table 1: Statistics

The combined dataset consists of context-question-answer triplets, which allows the model to learn to answer questions not just from the knowledge it already knows, but using the additional information provided.

5 Experiments

5.1 Metrics

We report Exact Match (EM) and token-level F1 as in SQuAD, plus BERTScore (Precision, Recall, F1) with xlm-roberta-large.

5.2 Experiment Setup

Hardware: 1×NVIDIA H100-80G for training; CPU deployment tested on Intel Xeon.

Retrieval: k=3; no reranker.

Training: 2 epochs, batch size 8, learning rate 2e-4, 16-bit bfloat16.

Decoding: two settings — (a) default sampling (T=0.7, top_p=0.8, top_k=20) and (b) deterministic (T=0, top_p=None, top_k=None).

5.3 Baselines

We used vanilla Qwen2.5-1.5B as a baseline

6 Results

Model	EM (%)	F1	BERT-P	BERT-R	BERT-F1
Qwen2.5-1.5B (a)	0.340	0.189	0.611	0.761	0.676
Qwen2.5-1.5B (b)	0.113	0.200	0.618	0.766	0.683
Qwen2.5-1.5B + LoRA (a)	30.0	0.492	0.723	0.738	0.730
Qwen2.5-1.5B + LoRA (b)	33.6	0.533	0.742	0.757	0.749

Table 2: Statistics

7 Conclusion

We presented a compact RAG assistant for Russian technical support. By adding $< 2\text{M}$ trainable parameters via LoRA, we multiplied exact-match accuracy on XQuAD while retaining a CPU-friendly footprint. Future work includes contrastive reranking, dynamic chunking, and continual training as documentation evolves.

References

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- [Lewis et al., 2020] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.